

Oliver Rupp

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Education

University of Notre Dame

B.S. in Physics, Honors Track; B.S. in Mathematics

Concentrations: Astrophysics and Advanced Physics

Notre Dame, IN

Expected May 2027

GPA: 4.0

Relevant Coursework: Quantum Mechanics I–II, Relativity: Special and General, Electromagnetism I–II, Modern Observational Techniques, Physics of Astrophysics, Advanced Mathematical Methods, Complex Variables, Abstract Algebra, Partial Differential Equations

Research Experience

University of Notre Dame

Research Assistant — Theoretical & Computational Astrophysics

Advisor: Dr. Grant Mathews

Notre Dame, IN

Spring 2024–Present

- Simulated core-collapse supernovae with the GR1D hydrodynamics code to investigate the effects of hadron–quark phase transitions on supernova evolution and explodability.
- Developed Python tools to convert hybrid hadron–quark equation-of-state (EOS) tables from CompOSE into the format required by GR1D, including interpolation, physical definitions, and unit conversions.
- Identified and resolved numerical instabilities in GR1D during the hadron–quark phase transition through dynamic Courant-factor reduction and implicit variable updates.
- Analyzed explodability and shock dynamics across EOS models; results presented at APS DNP CEU and incorporated into a manuscript in preparation.

University of Utah

NSF REU Researcher — Theoretical Astrophysics

Advisor: Dr. Tanmoy Laskar

Salt Lake City, UT

Summer 2026

- Benchmarked parameter recovery across three gamma-ray burst afterglow modeling codes: GAMMA, PyFRS, and VegasAfterglow.
- Developed a model-agnostic Python MCMC program to automate parameter sampling and analyze fitting degeneracies across afterglow models.
- Presented results at the Utah Summer Research Symposium.

University of Notre Dame

Research Assistant — Observational Astronomy

Advisor: Dr. Lauren Weiss

Notre Dame, IN

Fall 2026–Present

- Validate or demote transiting exoplanet candidates identified by NASA TESS, NASA Roman, and ESA PLATO using the Sarah L. Krizmanich Telescope.
- Analyze telescope observations using differential aperture photometry and MaxIm DL and AstroImage.

University of Notre Dame

Modern Observational Techniques Course - Observational Astronomy

Advisor: Dr. Jeffrey Chilcote

Notre Dame, IN

Fall 2025

- Observed M52 and NGC 884 using a 0.8 m telescope and determined the distance to NGC 884 through differential photometry and main-sequence matching.
- Measured the period of the SX Phoenicis variable CY Aquarii.

Selected Manuscript

Rupp, O., Mathews, G., & Boccioli, L. *Dynamics and Explodability of Core Collapse Supernovae with a Transition to Quark Matter*. Manuscript in preparation, 2026.

Presentations

- **Comparing Parameter Recovery Across GRB Afterglow Models** — Utah Summer Research Symposium, 2026.
- **Explodability of CCSNe with Transition to Quark Matter** — APS DNP CEU, 2025; Notre Dame Undergraduate Research Fair, 2025.
- **Mass Constraints on Primordial Black Holes** — Notre Dame Undergraduate Research Fair, 2024.

Technical Skills

Programming: Python (NumPy, SciPy, Matplotlib, AstroPy, emcee, h5py), Fortran, C++, Mathematica

Computing: Linux-based HPC clusters, Git/GitHub, LaTeX

Observational Astronomy: 0.8m telescope operation, MaxIm DL photometry software

Teaching & Leadership

Society of Physics Students

University of Notre Dame

Co-President

Fall 2026–Present

- Coordinate chapter meetings, delegate responsibilities across the executive board, and organize events including game nights, physics formal, and other community-building activities.
- Created a new undergraduate research presentation night to introduce underclassmen to research opportunities and encourage early involvement in physics research.
- Collaborate with the graduate outreach community on public outreach events connecting Notre Dame physics students with the broader community.

Cosmos Astronomy Club

University of Notre Dame

Co-President

Fall 2025–Present

- Organize and lead stargazing events featuring planets, stellar clusters, and other astronomical objects for the Notre Dame community.
- Volunteer with astronomy outreach events on campus and in the local community.

University of Notre Dame

Notre Dame, IN

Astronomy Teaching Assistant

Fall 2024–Present

- Lead nighttime telescope observations and teach students to identify and interpret astronomical objects using computerized telescopes.

Society of Physics Students

University of Notre Dame

Secretary

Fall 2025–Spring 2026

- Organized elections, outreach, mentoring, and social activities; improved connections between undergraduate and graduate students and led a graduate-school admissions workshop.

Awards & Honors

- APS DNP CEU Selected Participant, 2025
- Notre Dame College of Science Summer Undergraduate Research Fellowship, 2025
- Dean's List, Fall 2023–Present

Memberships

- American Physical Society
- Society of Physics Students
- Cosmos: Notre Dame Astronomy Club
- Notre Dame Folk Choir